Lab 7: Fuzz Testing with Hypothesis

Boarding bookings module — *pet_boarding.py* (repo L6_L7_49_pet_boarding)

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Lab	Lab 7 - Fuzzing for Input Validation Bugs
Tools	Python 3.13, pytest, Hypothesis

Summary

The three functions (*parse_booking*, *stay_average*, *payable_fee*) work for well-formed input but crash, or silently return nonsense, on unexpected input. Hypothesis fuzz tests exposed this: **13 of 16 tests failed** on the original code. After hardening the module, **all 16 tests pass**, including repeated runs of 3000 generated examples per test on three different random seeds.

Bugs found and how they were fixed

#	Bug (input -> original behaviour)	Fix
1	None / non-string input -> AttributeError (all 3 functions)	isinstance(x, str) guard; return None
2	Missing delimiter, e.g. "dog", "", "1500" -> ValueError (unpack)	Check len(split) == 2; return None
3	Extra delimiter, e.g. "dog:5:6" -> ValueError	Same length check
4	stay_average("2,,7"), "", "1.5,2", huge digit strings -> ValueError	Parse each item in try/except; return None
5	Empty pet name accepted: ":5" -> ('', 5.0)	Reject empty name after strip()
6	"dog:nan", "dog:inf", "dog:-5" accepted as valid days	_to_finite_float() + days >= 0 check
7	Negative stays accepted: "-5,-7" -> average -6	Reject negative values
8	payable_fee: "nan|1" -> nan, "1e999|0" -> inf, negative discount raised the fee	Reject non-finite / negative values; check overflow

Design decision: the functions never raise on bad input; they return **None** to signal "invalid", and *main()* prints a friendly message. This matches the "must not crash" property tested by Hypothesis.

Step 2 - Run the buggy application

```
python pet_boarding.py
```

```
Harshit Vaish | PES1UG24AM116 | Lab 7 - Fuzz Testing
$ python pet_boarding.py
==== Running parse_booking ====
Enter pet:days (e.g. dog:5): dog:5
Pet and Days: ('Dog', 5.0)

==== Running stay_average ====
Enter stay lengths in days as CSV (e.g. 2,5,7): 2,5,7
Average Stay: 4

==== Running payable_fee ====
Enter fee|discount (e.g. 1500|100): 1500|100
Payable Fee: 1670.0

$ python pet_boarding.py
==== Running parse_booking ====
Enter pet:days (e.g. dog:5): dog
Traceback (most recent call last):
  File "pet_boarding.py", line 31, in <module>
    main()
    ~~~~^^
  File "pet_boarding.py", line 17, in main
    result = parse_booking(user_input)
  File "pet_boarding.py", line 2, in parse_booking
    pet, days = text.split(':')
    ^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

Valid input works, but a malformed booking ("dog") crashes the program with a ValueError traceback.

Step 4 - Run the Hypothesis tests against the buggy code (summary)

```
python -m pytest test_pet_boarding.py
```

```
Harshit Vaish | PES1UG24AM116 | Lab 7 - Fuzz Testing
$ python -m pytest test_pet_boarding.py -q --tb=no -p no:cacheprovider
FFFFFFFFF.F.F.FF                                                         [100%]
========================== short test summary info ============================
FAILED test_pet_boarding.py::test_parse_booking_no_crash - ExceptionGroup: Hy...
FAILED test_pet_boarding.py::test_stay_average_no_crash - ExceptionGroup: Hyp...
FAILED test_pet_boarding.py::test_payable_fee_no_crash - ExceptionGroup: Hypo...
FAILED test_pet_boarding.py::test_parse_booking_any_type_no_crash - Exception...
FAILED test_pet_boarding.py::test_stay_average_any_type_no_crash - ExceptionG...
FAILED test_pet_boarding.py::test_payable_fee_any_type_no_crash - ExceptionGr...
FAILED test_pet_boarding.py::test_parse_booking_shaped_input_no_crash - Excep...
FAILED test_pet_boarding.py::test_stay_average_shaped_input_no_crash - ValueE...
FAILED test_pet_boarding.py::test_payable_fee_shaped_input_no_crash - Excepti...
FAILED test_pet_boarding.py::test_parse_booking_result_is_valid - ExceptionGr...
FAILED test_pet_boarding.py::test_stay_average_result_is_valid - ExceptionGro...
FAILED test_pet_boarding.py::test_payable_fee_result_is_valid - ExceptionGrou...
FAILED test_pet_boarding.py::test_regression_examples - ValueError: not enoug...
13 failed, 3 passed in 0.62s
```

13 failed, 3 passed: fuzzing breaks every function on the original code.

Step 4 (detail) - Counter-examples Hypothesis found

```
python -m pytest test_pet_boarding.py -k test_stay_average_no_crash
```

```
Harshit Vaish | PES1UG24AM116 | Lab 7 - Fuzz Testing
$ python -m pytest test_pet_boarding.py -q --tb=short -k test_stay_average_no_crash -p no:cacheprovider
F                                                                        [100%]
================================== FAILURES ===================================
_____________________________ test_stay_average_no_crash ______________________________
  + Exception Group Traceback (most recent call last):
  |   File "test_pet_boarding.py", line 37, in test_stay_average_no_crash
  |     def test_stay_average_no_crash(s):
  |                    ^^^
  |   File "/usr/local/lib/python.13/site-packages/hypothesis/core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "test_pet_boarding.py", line 39, in test_stay_average_no_crash
    |     stay_average(s)
    |     ~~~~~~~~~~~~^^^
    |   File "pet_boarding.py", line 6, in stay_average
    |     values = [int(r) for r in days_csv.split(',')]
    |               ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "test_pet_boarding.py", line 41, in test_stay_average_no_crash
    |     assert False, f"stay_average crashed with: {e!r}"
    | AssertionError: stay_average crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_stay_average_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "test_pet_boarding.py", line 39, in test_stay_average_no_crash
    |     stay_average(s)
    |     ~~~~~~~~~~~~^^^
    |   File "pet_boarding.py", line 6, in stay_average
    |     values = [int(r) for r in days_csv.split(',')]
    |                             ^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "test_pet_boarding.py", line 41, in test_stay_average_no_crash
    |     assert False, f"stay_average crashed with: {e!r}"
    | AssertionError: stay_average crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_stay_average_no_crash(
    |     s=None,
    | )
    +------------------------------------
========================== short test summary info ============================
FAILED test_pet_boarding.py::test_stay_average_no_crash - ExceptionGroup: Hyp...
1 failed, 15 deselected in 0.19s
```

Hypothesis shrinks failures to minimal inputs: s='' (ValueError) and s=None (AttributeError) for stay_average.

Step 5 - Run the fixed application

```
python pet_boarding.py
```

```
Harshit Vaish | PES1UG24AM116 | Lab 7 - Fuzz Testing
$ python pet_boarding.py
==== Running parse_booking ====
Enter pet:days (e.g. dog:5): dog:5
Pet and Days: ('Dog', 5.0)

==== Running stay_average ====
Enter stay lengths in days as CSV (e.g. 2,5,7): 2,5,7
Average Stay: 4

==== Running payable_fee ====
Enter fee|discount (e.g. 1500|100): 1500|100
Payable Fee: 1670.0

$ python pet_boarding.py
==== Running parse_booking ====
Enter pet:days (e.g. dog:5): dog
Pet and Days: Invalid input (expected pet:days, e.g. dog:5)

==== Running stay_average ====
Enter stay lengths in days as CSV (e.g. 2,5,7): 2,,7
Average Stay: Invalid input (expected whole numbers like 2,5,7)

==== Running payable_fee ====
Enter fee|discount (e.g. 1500|100): 1500
Payable Fee: Invalid input (expected fee|discount, e.g. 1500|100)
```

Valid input gives the same results as before; invalid input now produces a clear message instead of a crash.

Step 6 - Re-run the Hypothesis tests against the fixed code

```
python -m pytest test_pet_boarding.py -v
```

```
Harshit Vaish | PES1UG24AM116 | Lab 7 - Fuzz Testing
$ python -m pytest test_pet_boarding.py -v -p no:cacheprovider
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0 -- /usr/local/bin/python
hypothesis profile 'default'
rootdir: /home/claude/LAB7
plugins: platformdirs-4.13.5, anyio-4.15.1, hypothesis-6.170.15
collecting ... collected 16 items

test_pet_boarding.py::test_parse_booking_no_crash PASSED                 [  6%]
test_pet_boarding.py::test_stay_average_no_crash PASSED                  [ 12%]
test_pet_boarding.py::test_payable_fee_no_crash PASSED                   [ 18%]
test_pet_boarding.py::test_parse_booking_any_type_no_crash PASSED        [ 25%]
test_pet_boarding.py::test_stay_average_any_type_no_crash PASSED         [ 31%]
test_pet_boarding.py::test_payable_fee_any_type_no_crash PASSED          [ 37%]
test_pet_boarding.py::test_parse_booking_shaped_input_no_crash PASSED    [ 43%]
test_pet_boarding.py::test_stay_average_shaped_input_no_crash PASSED     [ 50%]
test_pet_boarding.py::test_payable_fee_shaped_input_no_crash PASSED      [ 56%]
test_pet_boarding.py::test_parse_booking_valid_roundtrip PASSED          [ 62%]
test_pet_boarding.py::test_parse_booking_result_is_valid PASSED          [ 68%]
test_pet_boarding.py::test_stay_average_valid_is_between_min_and_max PASSED [ 75%]
test_pet_boarding.py::test_stay_average_result_is_valid PASSED           [ 81%]
test_pet_boarding.py::test_payable_fee_valid_is_never_negative PASSED    [ 87%]
test_pet_boarding.py::test_payable_fee_result_is_valid PASSED            [ 93%]
test_pet_boarding.py::test_regression_examples PASSED                    [100%]

============================== 16 passed in 1.22s ==============================
```

All 16 tests pass.